

Similar Shapes

GCSE Maths · Worked Solutions

20 questions · 62 marks total

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Section A · Foundation

Questions 1–3 are the most straightforward; they get harder as you go on. Question 10 is the most challenging.

1 [1 mark]

Two similar triangles have corresponding sides of 3 cm and 9 cm.

Write down the scale factor of the enlargement.

1 Scale factor = $\frac{\text{new length}}{\text{original length}} = \frac{9}{3}$

2 $9 \div 3 = 3$, so every length in the larger triangle is 3 times as long.

Answer: 3

2 [2 marks]

Rectangle A measures 4 cm by 6 cm. Rectangle B measures 10 cm by 15 cm.

Show that the two rectangles are similar.

1 Compare the two pairs of corresponding sides.

2 $10 \div 4 = 2.5$ and $15 \div 6 = 2.5$

3 Both give the same scale factor, so the rectangles are similar.

Answer: Both ratios give a scale factor of 2.5, so they are similar

3 [2 marks]

Two similar triangles have corresponding sides 5 cm and 15 cm.

Another side of the smaller triangle is 7 cm. Find the corresponding side of the larger triangle.

1 Scale factor = $15 \div 5 = 3$

2 $7 \times 3 = 21$ cm

Answer: 21 cm

4 [2 marks]

A shape has perimeter 12 cm. It is enlarged by scale factor 3.

Write down the perimeter of the enlarged shape.

1 Perimeter is a length, so it is multiplied by the scale factor.

2 $12 \times 3 = 36$ cm

Answer: 36 cm

5

[3 marks]

Triangle A has sides 6 cm, 8 cm and 10 cm.

Triangle B is similar to triangle A and its shortest side is 9 cm.

Work out the other two sides of triangle B.

1 The shortest sides correspond: scale factor = $\frac{9}{6} = 1.5$

2 $8 \times 1.5 = 12$ cm

3 $10 \times 1.5 = 15$ cm

4 Check the ratios: 9 : 12 : 15 simplifies to 3 : 4 : 5, matching 6 : 8 : 10 ✓

Answer: 12 cm and 15 cm

6

[3 marks]

Two triangles are similar. The angles of the first are 40° , 60° and 80° .

(a) Write down the angles of the second triangle.

(b) Explain your answer.

1 (a) 40° , 60° and 80°

2 (b) Similar shapes are the same shape but a different size.

3 An enlargement changes lengths but never changes angles, so corresponding angles are equal.

Answer: (a) 40° , 60° and 80° (b) enlargement does not change angles

7

[3 marks]

A shape has area 20 cm^2 . It is enlarged by scale factor 2.

(a) Work out the area of the enlarged shape.

(b) Explain why the answer is not 40 cm^2 .

1 (a) Areas are multiplied by the **square** of the length scale factor.

2 $2^2 = 4$, so the new area is $20 \times 4 = 80 \text{ cm}^2$

3 (b) Area involves two lengths multiplied together, and each of those lengths doubles.

4 So the area is doubled twice over, i.e. multiplied by 4, not 2.

Answer: (a) 80 cm^2 (b) both dimensions double, so area is multiplied by 2^2

8

[3 marks]

In triangle ABC, the line DE is parallel to BC, with D on AB and E on AC.

AD = 4 cm, DB = 6 cm and AE = 6 cm.

Work out the length of EC.

1 Because DE is parallel to BC, triangles ADE and ABC are similar.

2 $AB = AD + DB = 4 + 6 = 10$ cm, so the scale factor from ADE to ABC is $\frac{10}{4} = 2.5$

3 $AC = 6 \times 2.5 = 15$ cm

4 $EC = AC - AE = 15 - 6 = 9$ cm

Answer: EC = 9 cm

9

[4 marks]

A photograph measures 8 cm by 12 cm.

It is enlarged so that the shorter side becomes 20 cm.

(a) Work out the scale factor.

(b) Work out the length of the longer side of the enlargement.

(c) Work out the area of the enlargement.

1 (a) Scale factor = $\frac{20}{8} = 2.5$

2 (b) $12 \times 2.5 = 30$ cm

3 (c) New area = $20 \times 30 = 600$ cm²

4 Check with the area scale factor: original area $8 \times 12 = 96$ cm² and $96 \times 2.5^2 = 96 \times 6.25 = 600$ ✓

Answer: (a) 2.5 (b) 30 cm (c) 600 cm²

10

[4 marks]

In triangle PQR, the line ST is parallel to QR, with S on PQ and T on PR.

PS = 5 cm, SQ = 3 cm and ST = 4 cm.

Work out the length of QR.

1 ST is parallel to QR, so triangles PST and PQR are similar (equal corresponding angles).

2 PQ = PS + SQ = 5 + 3 = 8 cm

3 Scale factor from PST to PQR = $\frac{PQ}{PS} = \frac{8}{5} = 1.6$

4 QR = ST $\times$ 1.6 = 4 $\times$ 1.6

5 = 6.4 cm

Answer: QR = 6.4 cm

Section B · Higher

Higher-tier style. Again, questions 1–3 are the gentlest and question 10 is the hardest.

1

[2 marks]

Two similar shapes have a length scale factor of 4.

Write down the area scale factor.

1 Area scale factor = (length scale factor)²

2 $4^2 = 16$

Answer: 16

2

[2 marks]

Two similar solids have a length scale factor of 3.

Write down the volume scale factor.

1 Volume scale factor = (length scale factor)³

2 $3^3 = 27$

Answer: 27

3

[3 marks]

Two similar shapes have areas 18 cm^2 and 50 cm^2 .

Work out the length scale factor.

1 Area scale factor = $\frac{50}{18} = \frac{25}{9}$

2 The length scale factor is the square root of the area scale factor.

3 $\sqrt{\left(\frac{25}{9}\right)} = \frac{5}{3}$

4 That is about 1.67.

Answer: $\frac{5}{3}$ (about 1.67)

4

[3 marks]

Two similar solids have volumes 24 cm^3 and 375 cm^3 .

Work out the length scale factor.

1 Volume scale factor = $\frac{375}{24} = 15.625$

2 The length scale factor is the cube root of the volume scale factor.

3 The cube root of 15.625 is 2.5, because $2.5^3 = 15.625$

Answer: 2.5

5

[4 marks]

Solid A has surface area 48 cm^2 and volume 32 cm^3 .

Solid B is similar to A and has surface area 108 cm^2 .

Work out the volume of solid B.

1 Area scale factor = $\frac{108}{48} = 2.25$

2 Length scale factor = $\sqrt{2.25} = 1.5$

3 Volume scale factor = $1.5^3 = 3.375$

4 Volume of B = 32×3.375

5 = 108 cm^3

Answer: 108 cm^3

6

[4 marks]

Two similar cones have heights 6 cm and 9 cm.

The smaller cone has volume $40\pi \text{ cm}^3$.

Work out the volume of the larger cone, in terms of π .

$$1 \quad \text{Length scale factor} = \frac{9}{6} = 1.5$$

$$2 \quad \text{Volume scale factor} = 1.5^3 = 3.375$$

$$3 \quad \text{Volume} = 40\pi \times 3.375$$

$$4 \quad = 135\pi \text{ cm}^3$$

Answer: $135\pi \text{ cm}^3$

7

[4 marks]

A cone of height 12 cm has volume 96 cm^3 .

The top of the cone is cut off by a cut parallel to the base, exactly halfway up.

(a) Explain why the small cone removed has volume 12 cm^3 .

(b) Work out the volume of the remaining frustum.

$$1 \quad \text{(a) The small cone is similar to the whole cone with length scale factor } \frac{1}{2}$$

$$2 \quad \text{So its volume scale factor is } \left(\frac{1}{2}\right)^3 = \frac{1}{8}$$

$$3 \quad \text{Volume} = 96 \times \frac{1}{8} = 12 \text{ cm}^3$$

$$4 \quad \text{(b) The frustum is what is left after removing the small cone.}$$

$$5 \quad 96 - 12 = 84 \text{ cm}^3$$

Answer: (a) volume scale factor is $\frac{1}{8}$ (b) 84 cm^3

8

[4 marks]

Triangle ABC is similar to triangle PQR.

AB = 6 cm, BC = 8 cm and PQ = 9 cm. The area of triangle ABC is 24 cm^2 .

(a) Work out the length of QR.

(b) Work out the area of triangle PQR.

$$1 \quad \text{(a) Length scale factor} = \frac{9}{6} = 1.5$$

$$2 \quad \text{QR} = 8 \times 1.5 = 12 \text{ cm}$$

$$3 \quad \text{(b) Area scale factor} = 1.5^2 = 2.25$$

$$4 \quad \text{Area of PQR} = 24 \times 2.25 = 54 \text{ cm}^2$$

Answer: (a) 12 cm (b) 54 cm^2

9

[4 marks]

Two similar statues are made from the same material.

The smaller statue is 30 cm tall and has mass 4.5 kg. The larger statue is 50 cm tall.

Work out the mass of the larger statue, to 1 decimal place.

$$1 \quad \text{Length scale factor} = \frac{50}{30} = \frac{5}{3}$$

2 Because the material is the same, mass is proportional to volume.

$$3 \quad \text{Volume (and mass) scale factor} = \left(\frac{5}{3}\right)^3 = \frac{125}{27}$$

$$4 \quad \text{Mass} = 4.5 \times \frac{125}{27} = \frac{562.5}{27} = 20.833\dots \text{ kg}$$

Answer: 20.8 kg (1 d.p.)

10

[5 marks]

In triangle ABC, angle ABC = 90°. BD is perpendicular to AC, with D on AC.

AD = 4 cm and DC = 9 cm.

(a) Explain why triangle ABD is similar to triangle BCD.

(b) Work out the length of BD.

(c) Work out the length of AB, giving your answer in surd form.

1 (a) Both triangles have a right angle at D.

2 Angle BAD and angle CBD are both equal to 90° – angle ABD, so they are equal.

3 Two pairs of equal angles means the triangles are similar.

$$4 \quad \text{(b) Matching corresponding sides: } \frac{AD}{BD} = \frac{BD}{DC}$$

$$5 \quad \text{So } BD^2 = AD \times DC = 4 \times 9 = 36, \text{ giving } BD = 6 \text{ cm}$$

$$6 \quad \text{(c) Triangle ABD is right-angled at D, so } AB^2 = AD^2 + BD^2$$

$$7 \quad AB^2 = 16 + 36 = 52, \text{ so } AB = \sqrt{52} = 2\sqrt{13} \text{ cm (about 7.21 cm)}$$

Answer: (b) BD = 6 cm (c) AB = 2√13 cm